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A Factorial Randomized Trial of Structured Information and Consultation Offers to Improve Comprehension of Executive Academic Programs Among Malaysian Private Sector Employees

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Abstract: This study tested whether structured, plain-language information and an offer of a brief consultation improved comprehension of an executive academic program among Malaysian private-sector employees and influenced subsequent information seeking. A 2 × 2 factorial randomized trial allocated 240 eligible employees (60 per group) to standard narrative or structured information, crossed with the presence or absence of a consultation offer. The primary outcome was a ten-item comprehension score assessed immediately after a 30-minute session. Analysis of covariance adjusted for baseline comprehension, with Holm adjustment for the two primary comparisons. The secondary outcome was a documented request for an application information package within seven days. Structured presentation increased baseline-adjusted comprehension by 0.68 points (95% CI 0.30 to 1.05; Holm-adjusted $p < 0.001$), and the consultation offer increased it by 0.59 points (95% CI 0.22 to 0.96; Holm-adjusted $p = 0.002$). The exploratory interaction was imprecisely estimated (0.60 points, 95% CI −0.16 to 1.36). The combined group had the highest adjusted mean (6.97 of 10 versus 5.70 in the control group). Consultation uptake was 64.2% among those offered it. No statistically significant effect of either intervention on seven-day information requests was detected (15.0% overall; risk differences of 0.05 and 0.00, respectively). These findings support clearer program information and accessible advising as ways to improve immediate understanding. Their effects on application, enrollment, and longer-term workforce or community outcomes remain to be established.

Keywords: Executive education; Lifelong learning; Adult learners; Information provision; Advising; Factorial randomized trial; Community capacity; Malaysia

1. Introduction

Technological change and labor market uncertainty have increased the importance of continuous skills development, with policy commentary describing the current decade as a period of large-scale upskilling (World Economic Forum, 2024). Malaysia's higher education policy positions lifelong learning as a pillar of national development. Executive academic programs provide pathways for employed adults to obtain diploma, bachelor's, and master's qualifications while remaining in employment (Ministry of Education Malaysia, 2015). Such programs combine qualifications with applied learning and accommodate adults who balance study, work, and family responsibilities (Ballou et al., 1999; Conger & Xin, 2000). Their relevance is reinforced by evidence linking employee human capital with organizational performance in the region (Arsyah & Pakri, 2024).

Access to adult learning can contribute to community development through organizational capability, household resilience, and knowledge exchange across professional and community networks. Prospective learners first need to understand a program's requirements, delivery arrangements, costs, and application procedures. Unclear or incomplete information can impede access even when programs are formally available.

1.1 Prior Evidence on Communication Preferences and Its Limits

A descriptive survey of 210 Malaysian private-sector employees examined executive education as a pathway for workforce capacity development and documented preferences for program information and engagement formats (Fakhrorazi et al., 2025). Respondents valued detailed curriculum information, flexibility, and personalized consultation. Social media was a prominent preferred information channel, consistent with research on digital channels in higher education choice (Constantinides & Zinck Stagno, 2012; Dwivedi et al., 2021). Stated preferences do not establish whether a communication practice improves understanding or changes behavior. An appealing format may not convey program requirements accurately, and valuing consultation does not establish its benefit. The present trial therefore extends the descriptive evidence by testing communication effects under randomized conditions.

1.2 Information, Advising, and Access to Adult Learning

Accounts of adult participation distinguish situational barriers, such as time, finances, and family responsibilities; institutional barriers, including program design, scheduling, procedures, and information quality; and dispositional barriers, such as confidence and perceived relevance (Cross, 1981). These barriers interact, allowing institutional communication and advising practices to amplify or offset other constraints (Boeren et al., 2010). From a capability perspective, an opportunity has limited practical value if an individual cannot understand it well enough to evaluate it (Sen, 2001).

College access experiments show that information and assistance can influence educational decisions. Helping families complete financial aid forms and providing personalized aid estimates increased applications and college enrollment, whereas information alone had no detectable effect (Bettinger et al., 2012). Customized information packets increased application and enrollment among high-achieving low-income students (Hoxby & Turner, 2013), while simplified, guaranteed cost information increased applications by reducing complexity and uncertainty (Dynarski et al., 2021). Brief information sessions and personalized outreach have also influenced aspirations and transitions, although effects depend on intervention design (Castleman & Page, 2015; Oreopoulos & Dunn, 2013). This evidence mainly concerns young people entering undergraduate study. Comparable experimental evidence for employed adults considering executive programs, particularly in Southeast Asia, remains scarce.

The interventions draw on two proposed mechanisms. Cognitive load theory suggests that headings, checklists, and sequenced steps make information easier to process than dense narrative text (Sweller, 1988), while information overload research links poor organization with lower decision quality (Eppler & Mengis, 2004). Adult learning theory emphasizes relating new information to personal circumstances (Knowles, 1988). A brief consultation may therefore help resolve program-specific uncertainties that generic materials cannot address. Qualitative evidence from Indonesian higher education similarly describes academic advisors as helping students interpret institutional rules when making academic decisions (Hidir et al., 2025). The logic model linking these communication conditions to the study outcomes is presented in Figure 1.

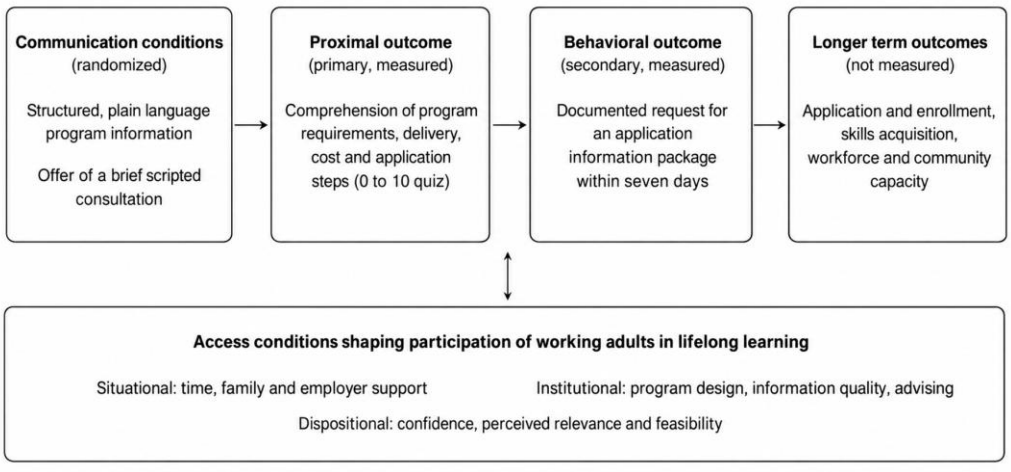


Figure 1. Logic model of communication conditions and outcomes

1.3 Objectives and Hypotheses

The primary objective was to estimate the effects of structured, plain-language information and a brief consultation offer on immediate comprehension, adjusting for baseline comprehension. H1 predicted higher

baseline-adjusted scores with structured information than with standard narrative information. H2 predicted higher baseline-adjusted scores with a consultation offer than without one. Both hypotheses were prespecified. The secondary objective was to assess effects on documented requests for an application information package within seven days. The interaction between presentation format and the consultation offer was exploratory. Consultation uptake and information requests were also described across the four groups.

The model links the communication conditions to the measured comprehension and information-request outcomes and to the longer-term workforce and community outcomes that motivated the study but were not measured. It situates the trial within community development research on access to learning opportunities. Arrows denote proposed pathways, including relationships not tested here.

2. Methodology

2.1 Design

The study was an individually randomized 2 × 2 factorial trial with equal allocation to four groups (Montgomery et al., 2003). Factor A compared standard narrative information (A0) with structured, plain-language information using headings, a requirements checklist, a comparison checklist, and a sequential application guide (A1). Factor B compared self-directed reading without a consultation offer (B0) with an offer of an optional scripted consultation lasting up to ten minutes (B1). The groups were A0B0 (control), A1B0, A0B1, and A1B1. The trial is presented with reference to CONSORT 2010 and its extension for factorial randomized trials (Kahan et al., 2023; Schulz et al., 2010). The hypotheses, outcomes, and analysis plan were specified before recruitment. The trial was not registered in a public trial registry.

2.2 Setting, Participants, and Recruitment

Participants were recruited in Malaysia through workplaces and professional networks using purposive, nonprobability recruitment during 2025, following the author team's earlier survey work. Standardized recruitment materials emphasized voluntary participation. Eligible participants were aged 18 years or older, employed in the Malaysian private sector, had at least two years of work experience, were considering further academic study for professional development, met the selected program's published entry prerequisites, could complete the study in the language of the approved materials, and provided informed consent. The two-year experience criterion was a study condition rather than a universal admission requirement. Exclusions covered duplicate registration, participation in the instrument pilot, current enrollment in the selected program, prior exposure to the intervention materials, and self-reported participation in the author team's earlier executive-education surveys. A screening log recorded eligibility, participation decisions, and randomization, with exclusion reasons coded in predefined categories. Random assignment supports between-group comparisons but does not make the sample representative of Malaysian private-sector employees.

2.3 Program Selection and Intervention Materials

The intervention concerned one executive master's program offered by a Malaysian public university. The program is not named to protect the identities of the institution and supporting staff. Its qualification level, admission requirements, curriculum, delivery arrangements, duration, fees, assessment requirements, and application procedures were verified against official documentation before the materials were finalized. Item-by-item checks confirmed that both versions contained identical facts, and their lengths were approximately equal. The standard version used conventional brochure-style paragraphs; the structured version used plain-language headings, checklists, and a numbered application sequence. Because structure and wording changed together, Factor A estimates the effect of the combined presentation package rather than its individual elements.

Participants assigned to B1 were offered a consultation delivered by trained facilitators using a standardized script. Facilitators clarified existing information and were instructed not to introduce additional promotional claims. Attendance, duration, and question categories were recorded. All participants had the same 30-minute intervention window; accepted consultations took place within that window and before post-intervention assessment. Participants who declined remained in their assigned group, so the randomized comparison estimates the effect of the offer.

2.4 Procedure, Randomization, and Blinding

Sessions proceeded through eligibility screening and informed consent, collection of baseline characteristics, a common neutral program primer, baseline comprehension assessment, concealed random allocation, a 30-minute intervention session, and immediate post-intervention comprehension assessment. All groups then had identical

opportunities to request an application information package over seven days. Because the primer preceded baseline assessment in every group, the trial estimates the additional benefit of the assigned communication conditions beyond a shared initial exposure.

An independent researcher not involved in recruitment generated the allocation sequence using randomly varying block sizes compatible with equal allocation. Allocation remained concealed until baseline assessment was complete. Participants and facilitators could not be blinded. Quiz scoring was automated against a fixed answer key without access to allocation. Individual sessions and instructions not to share materials during follow-up were used to limit contamination.

2.5 Outcomes and Measures

The primary outcome was immediate post-intervention comprehension, adjusted for baseline comprehension. A study-specific ten-item assessment covered entry requirements, credit load and duration, delivery arrangements, fees and payment, assessment requirements, and application procedures. Each correct answer received one point, giving a total score from 0 to 10. Baseline and post-intervention forms underwent content review by academics experienced in adult and executive education and were piloted with working adults who did not participate in the main trial. Items, answer keys, scoring rules, and material versions were finalized before recruitment. The assessment measured factual understanding of the selected program rather than general knowledge of executive education.

The secondary outcome was a documented, time-stamped request for an application information package within seven days of the session. All groups received the same request mechanism, instructions, and follow-up opportunity; zero indicated no request during complete monitoring. This measured information seeking rather than application or enrollment. Baseline variables were age, education (diploma, bachelor’s degree, or postgraduate qualification), work experience, digital confidence (single-item, 1 to 5), and perceived time constraint (single-item, 1 to 5). The single-item ratings were descriptive and exploratory measures rather than validated constructs. Consultation uptake and duration were recorded as implementation measures.

2.6 Sample Size

The sample size was calculated prospectively for the two primary main effect comparisons. Assuming a smallest meaningful difference of 0.7 points on the 10-point scale, a standard deviation of 2.0, a baseline-to-follow-up correlation of 0.6, a two-sided significance level of 0.025 for each primary comparison (the most conservative Holm threshold), and 80% power, an analysis of covariance required approximately 100 participants per level of each factor. Allowing for up to 15% missing outcomes and balanced blocks, the target was set at 240 participants (120 per factor level, 60 per group). The interaction analysis was not powered and was treated as exploratory.

2.7 Statistical Analysis

The primary analysis followed the intention-to-treat principle, retaining participants in their assigned groups regardless of consultation uptake. Analysis of covariance adjusted post-intervention comprehension for baseline comprehension, a prespecified approach to improve precision (Vickers & Altman, 2001). With the factors coded as centered indicators, the model was as follows:

$$\mathrm{Post\ score} = \beta_0 + \beta_1(\mathrm{Baseline\ score}) + \beta_2(A - 0.5) + \beta_3(B - 0.5) + \beta_4\left[(A - 0.5)(B - 0.5)\right] + \varepsilon \tag{1}$$

where, A indicates structured presentation and B indicates the consultation offer. Under this coding, β_1 is the common slope for baseline comprehension, β_2 is the presentation effect averaged equally across consultation conditions, β_3 is the consultation-offer effect averaged equally across presentation conditions, and β_4 is the interaction, expressed as the difference between the presentation effects in the two consultation conditions. Adjusted means for the four groups were evaluated at the sample mean baseline score. The two primary comparisons used two-sided tests with Holm adjustment to control the familywise error rate at 0.05 (Holm, 1979). The 95% confidence intervals are pointwise and are not adjusted for multiplicity. The interaction was exploratory, and a nonsignificant interaction is not interpreted as evidence that the components act independently (Kahan, 2013). Heteroskedasticity-robust (HC3) standard errors were used throughout. Employer-level clustering was not modeled.

Model assumptions, ceiling effects, and residual distributions were examined. Four prespecified sensitivity analyses used a change-score model, rank-based analysis of covariance, a binomial generalized linear model treating the score as correct answers out of ten, and the primary model additionally adjusted for age, education, work experience, digital confidence, and time constraint. A nonparametric bootstrap (2,000 resamples) provided

additional confidence intervals. The secondary outcome was summarized by group using request counts, denominators, risk differences, and 95% confidence intervals, supplemented by logistic regression with the same factorial coding. The planned approach allowed for multiple imputation if outcomes were missing (White et al., 2011); complete outcome data made imputation unnecessary. Consultation uptake and other post-randomization measures were excluded from the primary model's covariates. Analyses used Python 3.12, statsmodels 0.14, and SciPy.

3. Results

3.1 Participant Flow and Baseline Characteristics

Figure 2 summarizes participant flow from screening through analysis. Of 300 individuals assessed for eligibility, 60 were excluded, namely 30 who did not meet the eligibility criteria, 20 who declined to participate, and 10 for other reasons. The remaining 240 participants completed baseline assessment and were randomized, 60 to each group. All 240 completed the 30-minute session and the post-intervention assessment, and seven-day monitoring of information requests was complete for all participants, so no outcome data were missing and multiple imputation was not required. Among the 120 participants offered a consultation, 77 (64.2%) accepted it, namely 40 of 60 (66.7%) in A0B1 and 37 of 60 (61.7%) in A1B1.

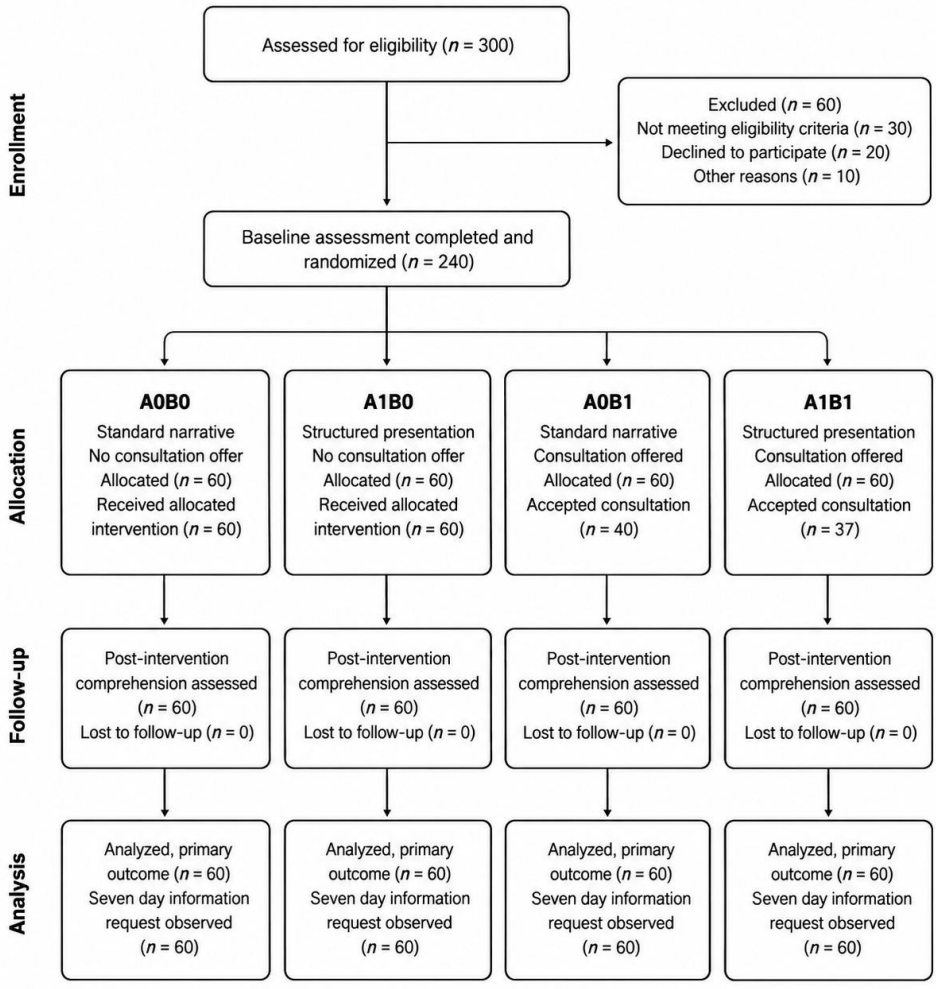


Figure 2. Participant flow through the factorial trial

Participants had a mean age of 38.8 years and a mean of 10.6 years of work experience (Table 1). Overall, 60.4% held a bachelor's degree, 24.2% a diploma, and 15.4% a postgraduate qualification. Mean baseline comprehension was 4.77 of 10 (SD 1.63), indicating that participants answered approximately half of the questions correctly after the common primer. Groups were broadly comparable on observed characteristics, with some differences in age, experience, and baseline comprehension. Baseline comprehension was included in the primary model, and the

other variables were examined in sensitivity analysis.

Table 1. Baseline characteristics by randomized group

Characteristic	A0B0 (*n* = 60)	A1B0 (*n* = 60)	A0B1 (*n* = 60)	A1B1 (*n* = 60)	Total (*n* = 240)
Age, years, mean (SD)	38.1 (10.6)	41.2 (9.3)	38.7 (10.1)	37.1 (9.4)	38.8 (9.9)
Work experience, years, mean (SD)	10.1 (8.0)	12.6 (9.3)	10.8 (8.3)	8.8 (6.6)	10.6 (8.1)
Education, *n* (%)					
Diploma	15 (25.0)	16 (26.7)	11 (18.3)	16 (26.7)	58 (24.2)
Bachelor's degree	31 (51.7)	34 (56.7)	42 (70.0)	38 (63.3)	145 (60.4)
Postgraduate	14 (23.3)	10 (16.7)	7 (11.7)	6 (10.0)	37 (15.4)
Digital confidence (1 to 5), mean (SD)	3.38 (0.87)	3.30 (0.91)	3.20 (0.84)	3.13 (0.93)	3.25 (0.89)
Perceived time constraint (1 to 5), mean (SD)	3.15 (0.95)	2.95 (1.03)	2.83 (0.94)	3.40 (1.01)	3.08 (1.00)
Baseline comprehension (0 to 10), mean (SD)	4.57 (1.56)	5.03 (1.66)	4.93 (1.49)	4.55 (1.78)	4.77 (1.63)

Note: A0 = standard narrative; A1 = structured presentation; B0 = no consultation offer; B1 = consultation offered; SD = standard deviation. Significance tests of baseline differences are not reported, consistent with CONSORT guidance.

3.2 Primary Outcome: Comprehension

Mean post-intervention scores (Table 2 and Figure 3) were 5.50 in the control group (A0B0), 6.33 with structured presentation alone (A1B0), 6.15 with the consultation offer alone (A0B1), and 6.75 with both components (A1B1). Baseline comprehension was strongly associated with post-intervention comprehension (coefficient 0.98, 95% CI 0.87 to 1.10; correlation 0.72), and the primary model explained 57% of the variance in post-intervention scores. Baseline adjustment preserved the ordering of the group means and slightly widened the gap between the combined and control groups. Adjusted means were 5.70 (A0B0), 6.08 (A1B0), 5.99 (A0B1), and 6.97 (A1B1).

Table 2. Comprehension scores (0 to 10) by randomized group, *n* = 60 per group

Group	Baseline Mean (SD)	Post Mean (SD)	Post Median	Change Mean (SD)	Adjusted Post Mean (95% CI)
A0B0 Standard, no offer	4.57 (1.56)	5.50 (2.09)	5.5	0.93 (1.31)	5.70 (5.36 to 6.04)
A1B0 Structured, no offer	5.03 (1.66)	6.33 (2.41)	6.0	1.30 (1.55)	6.08 (5.67 to 6.48)
A0B1 Standard, offer	4.93 (1.49)	6.15 (2.10)	6.5	1.22 (1.54)	5.99 (5.59 to 6.39)
A1B1 Structured, offer	4.55 (1.78)	6.75 (2.04)	7.0	2.20 (1.40)	6.97 (6.61 to 7.33)

Note: A0 = standard narrative; A1 = structured presentation; B0 = no consultation offer; B1 = consultation offered. Adjusted means are from the primary analysis of covariance evaluated at the sample mean baseline score (4.77). CI = confidence interval; SD = standard deviation.

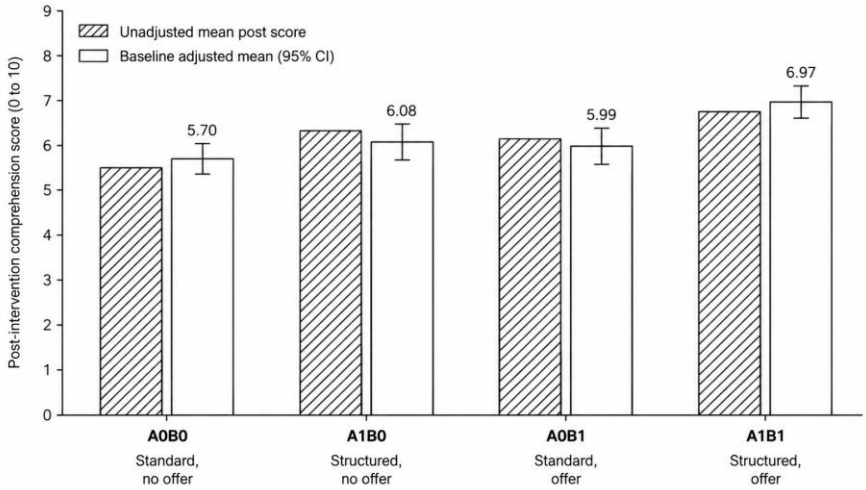


Figure 3. Unadjusted and baseline-adjusted mean comprehension scores by group

Averaged across consultation conditions, structured presentation increased baseline-adjusted comprehension by 0.68 points (95% CI 0.30 to 1.05; $p < 0.001$; Holm-adjusted $p < 0.001$), supporting H1 (Table 3). Averaged across

presentation conditions, the consultation offer increased comprehension by 0.59 points (95% CI 0.22 to 0.96; p = 0.002; Holm-adjusted p = 0.002), supporting H2. Both effects remained statistically significant after Holm adjustment at the 0.05 familywise level. Expressed relative to the standard deviation of the post-intervention score (2.20), these correspond to standardized effects of 0.31 and 0.27, and relative to the residual standard deviation of the adjusted model (1.46) to 0.46 and 0.40.

Table 3. Primary analysis of covariance for post-intervention comprehension (n = 240)

Estimate	Points (95% CI)	*p* Value	Holm-Adjusted *p*
Baseline comprehension (per point)	0.98 (0.87 to 1.10)	<0.001	–
Structured presentation, main effect (H1)	0.68 (0.30 to 1.05)	<0.001	<0.001
Consultation offer, main effect (H2)	0.59 (0.22 to 0.96)	0.002	0.002
Presentation × consultation interaction (exploratory)	0.60 (−0.16 to 1.36)	0.121	–
Simple effect of presentation, no consultation offer	0.33 (−0.20 to 0.86)	0.226	–
Simple effect of presentation, consultation offered	0.94 (0.40 to 1.48)	<0.001	–
Simple effect of consultation offer, standard narrative	0.28 (−0.25 to 0.82)	0.297	–
Simple effect of consultation offer, structured presentation	0.89 (0.34 to 1.44)	0.002	–

Note: CI = confidence interval; "–" = not applicable. CIs are unadjusted for multiplicity. Holm adjustment applies only to H1 and H2. Simple effects are descriptive estimates from separate baseline-adjusted models.

The exploratory interaction was 0.60 points (95% CI −0.16 to 1.36; p = 0.121). Separate baseline-adjusted models fitted within each level of the other factor suggested a larger presentation effect when consultation was offered (0.94 points, 95% CI 0.40 to 1.48) than when it was not (0.33 points, 95% CI −0.20 to 0.86). The corresponding consultation-offer effects were 0.89 points (95% CI 0.34 to 1.44) with structured materials and 0.28 points (95% CI −0.25 to 0.82) with narrative materials. These estimates are descriptive and were obtained from separately fitted models; they are not exact decompositions of the primary-model interaction. The trial was not powered to detect interaction, and the wide confidence interval neither establishes nor rules out synergy between the components.

3.3 Assumption Checks and Sensitivity Analyses

Residuals showed no evidence of heteroskedasticity (Breusch–Pagan p = 0.56) and only mild departure from normality (absolute skewness 0.19, kurtosis 3.0). HC3 standard errors were used to reduce sensitivity to variance misspecification. Ceiling effects were limited: 18 participants (7.5%) scored 10, and 36 (15.0%) scored 9 or 10. The primary conclusions were consistent across the prespecified sensitivity analyses (Table 4). The change-score model produced main effects of 0.68 and 0.59 points, closely matching the primary model. The rank-based analysis of covariance and binomial generalized linear model both yielded statistically significant main effects ($p \leq 0.002$). Additional adjustment for baseline covariates changed the estimates little (0.67 and 0.55 points). Bootstrap percentile intervals closely matched the analytic intervals. Interaction estimates in the change-score, binomial, and additional-covariate models were positive but imprecise (p = 0.076 to 0.340). The unadjusted post-intervention model, which was not a prespecified sensitivity analysis, gave an interaction estimate of −0.23 (p = 0.679).

Table 4. Sensitivity analyses for the primary outcome

Model	Presentation (A)	Consultation Offer (B)	Interaction (A × B)
Primary ANCOVA, HC3 (reference)	0.68 (0.30 to 1.05), p < 0.001	0.59 (0.22 to 0.96), p = 0.002	0.60 (−0.16 to 1.36), p = 0.121
Unadjusted difference in post-intervention scores	0.72 (0.17 to 1.27), p = 0.011	0.53 (−0.02 to 1.08), p = 0.058	−0.23 (−1.34 to 0.87), p = 0.679
Change-score model	0.68, p < 0.001	0.59, p = 0.002	0.62, p = 0.104
Rank-based ANCOVA	p < 0.001	p = 0.002	p = 0.134
Binomial GLM (logit), coefficient	0.34, p < 0.001	0.28, p = 0.002	0.32, p = 0.076
ANCOVA with baseline covariates added	0.67 (0.29 to 1.05), p < 0.001	0.55 (0.15 to 0.94), p = 0.007	0.39 (−0.41 to 1.19), p = 0.340
Bootstrap percentile 95% CI (2,000 resamples)	0.30 to 1.04	0.20 to 0.96	−0.12 to 1.33

Note: Effects are in score points, except binomial GLM coefficients (log odds); parentheses indicate 95% CIs. ANCOVA = analysis of covariance; GLM = generalized linear model; CI = confidence interval.

3.4 Consultation Uptake

Among those offered a consultation, uptake was 64.2% (77 of 120): 66.7% with narrative materials and 61.7% with structured materials. Accepted consultations remained within the ten-minute limit, and facilitator logs

indicated that questions most often concerned fees and payment arrangements, scheduling and delivery mode, and admission requirements. Within the offer groups, participants who accepted consultation had higher mean post-intervention scores than those who declined (6.52 versus 5.40 in A0B1 and 7.00 versus 6.35 in A1B1), despite similar baseline scores. Because consultation uptake was self-selected, these comparisons are observational and describe implementation rather than the causal effect of attending consultation. The randomized estimate is the effect of the offer.

3.5 Secondary Outcome: Information Requests within Seven Days

Overall, 36 of 240 participants (15.0%) requested the application information package within seven days (Table 5). Request rates were 11.7% in A0B0, 18.3% in A1B0, 13.3% in A0B1, and 16.7% in A1B1 (overall chi-square $p = 0.727$). The risk difference was 0.05 for structured presentation versus standard information (95% CI −0.04 to 0.14; $p = 0.366$) and 0.00 for the consultation offer versus no offer (95% CI −0.09 to 0.09; $p = 1.000$). The logistic model yielded odds ratios of 1.49 (95% CI 0.72 to 3.05) for presentation and 1.02 (95% CI 0.50 to 2.09) for the offer, with no statistically significant interaction detected. Mean post-intervention scores were 6.39 among participants who requested the package and 6.15 among those who did not. No statistically significant effect of either intervention on short-term information seeking was detected. However, the trial was not powered for this binary outcome, and the confidence intervals allow both decreases and increases in request rates.

Table 5. Documented requests for an application information package within seven days

Comparison	Requests/*n* (%)	Risk Difference (95% CI)	*p* Value
	By group		
A0B0 Standard, no offer	7/60 (11.7)	Reference	–
A1B0 Structured, no offer	11/60 (18.3)	0.07 (−0.06 to 0.19)	–
A0B1 Standard, offer	8/60 (13.3)	0.02 (−0.10 to 0.13)	–
A1B1 Structured, offer	10/60 (16.7)	0.05 (−0.07 to 0.17)	0.727 (overall)
	By factor		
Structured ($n = 120$) vs. standard ($n = 120$)	21 (17.5) vs. 15 (12.5)	0.05 (−0.04 to 0.14)	0.366
Offer ($n = 120$) vs. no offer ($n = 120$)	18 (15.0) vs. 18 (15.0)	0.00 (−0.09 to 0.09)	1.000

Note: A0 = standard narrative; A1 = structured presentation; B0 = no consultation offer; B1 = consultation offered. CI = confidence interval; "–" = not applicable. The four-group comparison used Pearson's chi-square test; factor-level comparisons used continuity-corrected chi-square tests. Fisher's exact tests gave consistent conclusions. Risk differences are expressed as proportions.

4. Discussion

4.1 Principal Findings

Both communication practices improved immediate comprehension in this sample. Structured, plain-language information increased baseline-adjusted comprehension by approximately two-thirds of a point on a 10-point scale, and the consultation offer by approximately six-tenths of a point, each averaged across the other factor. The combined group's adjusted mean exceeded the control mean by 1.27 points, corresponding to approximately 70% versus 57% correct responses. The positive but imprecise interaction estimate leaves synergy uncertain. Neither intervention had a statistically significant effect on seven-day information requests.

4.2 Interpretation in Relation to Prior Work

The trial adds experimental evidence to the descriptive findings of Fakhrorazi et al. (2025), who reported preferences for detailed information, flexibility, and one-to-one consultation. The findings are consistent with cognitive load accounts of organized information processing (Eppler & Mengis, 2004; Sweller, 1988) and with adult learning theory's emphasis on relating information to individual circumstances (Knowles, 1988).

Neither communication intervention produced a statistically significant change in information requests. College access studies have found behavioral benefits from application assistance and reduced complexity or uncertainty (Bettinger et al., 2012; Dynarski et al., 2021; Hoxby & Turner, 2013). The brief consultation tested here clarified information without helping participants apply, secure financing, or negotiate employer support. Time, cost, employer flexibility, and administrative burden may affect subsequent decisions (Boeren et al., 2010; Cross, 1981), but these mechanisms were not tested. Seven days is also a short period for such decisions, and requesting information is only an early behavioral response. Effects on application or enrollment therefore remain unknown.

Baseline comprehension remained strongly associated with post-intervention comprehension after adjustment for the communication conditions. This association does not show whether either intervention narrowed gaps in understanding. Further research should examine whether benefits vary with prior knowledge and whether targeted advising helps less-informed learners.

4.3 Implications for Community Development Practice and Policy

A mean baseline score below five correct answers out of ten indicates incomplete understanding even after a common primer. This illustrates a potential gap between formal availability of learning opportunities and usable access. Financial inclusion research makes a related argument that access needs to be accompanied by literacy (Grohmann et al., 2018) and that institutions can help develop it (Asetya & Dahlan, 2026). For universities, the findings support presenting requirements, time commitments, schedules, fees, and application steps as decision support. Clear communication is also relevant to flexible and blended programs (Anthony et al., 2022). Research on blended delivery and achievement provides related context (Tom et al., 2026), although it does not establish the effects of the communication practices tested here.

Employers could facilitate brief advising through workplace learning sessions to help employees understand study options, supporting broader links between learning and career development (Bridgstock et al., 2019). For lifelong learning policy (Ministry of Education Malaysia, 2015), information standards and advising warrant consideration. The absence of a statistically significant short-term behavioral effect, however, cautions against assuming that clearer information increases participation. Application assistance, employer time release, and financing pathways warrant evaluation alongside information improvements. Wider organizational and community benefits discussed in executive education research depend on subsequent participation and learning, which were not measured here (Conger & Xin, 2000; Crotty & Soule, 1997; Myrsiades, 2001).

4.4 Strengths and Limitations

Strengths include concealed individual randomization, prespecified baseline adjustment and multiplicity control, objective scoring without access to allocation, complete outcome data, and efficient estimation of two interventions in one sample. Sensitivity analyses supported the primary conclusions. Purposive recruitment limits generalizability, while the single executive master's program limits transferability across qualifications and institutions. Only immediate comprehension was assessed, so longer-term retention is unknown. Repeated testing and any differences in difficulty between the baseline and post-intervention quizzes may also have influenced changes in scores. The quiz measured factual understanding of one program and did not directly assess the ability to make an informed enrollment decision. Structure and wording changed together, while the consultation offer combined clarification with human attention, preventing isolation of their individual contributions. Workplace contamination or dependence could not be excluded, and employer-level clustering was not modeled. The trial was not powered for the secondary outcome, which captured information seeking rather than participation. Digital confidence and time constraint were descriptive single-item ratings.

5. Conclusions

Structured, plain-language information and a brief consultation offer improved immediate comprehension among the employees recruited to this trial. The combined group had the highest adjusted mean, but neither intervention had a statistically significant effect on seven-day information requests. Clearer information and accessible advising may support understanding; their effects on application, enrollment, and longer-term workforce or community outcomes remain unknown. Future studies should evaluate application assistance and employer support, extend follow-up to application and enrollment, examine benefits by baseline comprehension, and replicate the trial across programs, institutions, and populations.

Author Contributions

Conceptualization, H.H. and A.F.; methodology, H.H. and R.B.I.; investigation and data curation, H.H.; formal analysis, R.B.I. and H.H.; writing—original draft preparation, H.H.; writing—review and editing, A.F. and R.B.I.; visualization, R.B.I.; supervision and project administration, H.H.; funding acquisition, H.H. All authors have read and approved the final manuscript.

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Informed Consent Statement

All participants provided written informed consent before any research assessment, after being informed about study procedures, voluntary participation, withdrawal, foreseeable burdens, confidentiality, and data use. Neither individual quiz scores nor employees' participation decisions were disclosed to employers.

Ethical Approval

The study protocol was reviewed and approved by the institutional research ethics committee of Universiti Utara Malaysia before participant recruitment.

Data Availability

The participant-level dataset, intervention materials, comprehension instruments and answer keys, and analysis code are available from the corresponding author on reasonable request for academic purposes. Follow-up contact details were stored separately and are not shared. Data from the author team's earlier surveys were not imported into the trial dataset, and previous participation was screened at enrollment.

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Conflicts of Interest

The authors declare no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

Generative AI tools were not used to generate, analyze, or interpret the data or results reported in this study. The authors are responsible for the manuscript's accuracy, originality, and compliance with ethical and publishing standards.

References

Anthony, B., Kamaludin, A., Romli, A., Raffei, A. F. M., Phon, D. N. A. L. E., Abdullah, A., & Ming, G. L. (2022). Blended learning adoption and implementation in higher education: A theoretical and systematic review. *Technol. Knowl. Learn.*, *27*, 531–578. https://doi.org/10.1007/s10758-020-09477-z.

Arsyah, T. D. & Pakri, P. (2024). Leveraging human capital for performance enhancement in Indonesia technology sector. *J. Econ. Bus. Lett.*, *4*(3), 12–20. https://doi.org/10.55942/jebl.v4i3.323.

Asetya, D. R. & Dahlan, S. P. (2026). Islamic financial literacy: Determinants, measurement, and outcomes. *J. Islam. Econ. Insights*, *2*(1), 15–29. https://doi.org/10.55942/jiei.v2i1.1742.

Ballou, R., Bowers, D., Boyatzis, R. E., & Kolb, D. A. (1999). Fellowship in lifelong learning: An executive development program for advanced professionals. *J. Manag. Educ.*, *23*(4), 338–354. https://doi.org/10.1177/105256299902300403.

Bettinger, E. P., Long, B. T., Oreopoulos, P., & Sanbonmatsu, L. (2012). The role of application assistance and information in college decisions: Results from the H&R Block FAFSA experiment. *Q. J. Econ.*, *127*(3), 1205–1242. https://doi.org/10.1093/qje/qjs017.

Boeren, E., Nicaise, I., & Baert, H. (2010). Theoretical models of participation in adult education: The need for an integrated model. *Int. J. Lifelong Educ.*, *29*(1), 45–61. https://doi.org/10.1080/02601370903471270.

Bridgstock, R., Grant-Iramu, M., & McAlpine, A. (2019). Integrating career development learning into the curriculum: Collaboration with the careers service for employability. *J. Teach. Learn. Grad. Employab.*, *10*(1), 56–72. https://doi.org/10.21153/jtlge2019vol10no1art785.

Castleman, B. L. & Page, L. C. (2015). Summer nudging: Can personalized text messages and peer mentor outreach increase college going among low-income high school graduates? *J. Econ. Behav. Organ.*, *115*, 144–160. https://doi.org/10.1016/j.jebo.2014.12.008.

Conger, J. A. & Xin, K. (2000). Executive education in the 21st century. *J. Manag. Educ.*, *24*(1), 73–101. https://doi.org/10.1177/105256290002400106.

Constantinides, E. & Zinck Stagno, M. C. (2012). Higher education marketing: A study on the impact of social media on study selection and university choice. *Int. J. Technol. Educ. Mark.*, *2*(1), 41–58. https://doi.org/10.4018/ijtem.2012010104.

Cross, K. P. (1981). *Adults as Learners: Increasing Participation and Facilitating Learning*. Jossey-Bass.

Crotty, P. T. & Soule, A. J. (1997). Executive education: Yesterday and today, with a look at tomorrow. *J. Manag. Dev.*, *16*(1), 4–21. https://doi.org/10.1108/02621719710155445.

Dwivedi, Y. K., Ismagilova, E., Hughes, D. L., Carlson, J., Filieri, R., Jacobson, J., Jain, V., Karjaluoto, H., Kefi,

H., Krishen, A. S., et al. (2021). Setting the future of digital and social media marketing research: Perspectives and research propositions. *Int. J. Inf. Manag.*, *59*, 102168. https://doi.org/10.1016/j.ijinfomgt.2020.102168.

Dynarski, S., Libassi, C. J., Michelmore, K., & Owen, S. (2021). Closing the gap: The effect of reducing complexity and uncertainty in college pricing on the choices of low-income students. *Am. Econ. Rev.*, *111*(6), 1721–1756. https://doi.org/10.1257/aer.20200451.

Eppler, M. J. & Mengis, J. (2004). The concept of information overload: A review of literature from organization science, accounting, marketing, MIS, and related disciplines. *Inf. Soc.*, *20*(5), 325–344. https://doi.org/10.1080/01972240490507974.

Fakhrorazi, A., Hartini, H., Omar, R., & Ikhsan, R. B. (2025). Executive education as a tool for workforce capacity development: Evidence from Malaysian private sector employees. *Cent. Community Dev. J.*, *5*(2), 20–40. https://doi.org/10.55942/ccdj.v5i2.1046.

Grohmann, A., Klühs, T., & Menkhoff, L. (2018). Does financial literacy improve financial inclusion? Cross country evidence. *World Dev.*, *111*, 84–96. https://doi.org/10.1016/j.worlddev.2018.06.020.

Hidir, A., Gea, Y. K., Rusmawati, P., Fajri, A. H. G., Vitamagistra, S. W., & Mardatillah, M. (2025). The role of academic advisors in assisting sociology students Faculty of Social and Political Sciences, University of Riau. *Priviet Soc. Sci. J.*, *5*(10), 106–113. https://doi.org/10.55942/pssj.v5i10.631.

Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scand. J. Stat.*, *6*(2), 65–70.

Hoxby, C. & Turner, S. (2013). *Expanding college opportunities for high-achieving, low-income students* (SIEPR Discussion Paper No. 12-014). Stanford Institute for Economic Policy Research. https://siepr.stanford.edu/publications/working-paper/expanding-college-opportunities-high-achieving-low-income-students

Kahan, B. C. (2013). Bias in randomised factorial trials. *Stat. Med.*, *32*(26), 4540–4549. https://doi.org/10.1002/sim.5869.

Kahan, B. C., Hall, S. S., Beller, E. M., Birchenall, M., Chan, A.-W., Elbourne, D., Little, P., Fletcher, J., Golub, R. M., Goulao, B., et al. (2023). Reporting of factorial randomized trials: Extension of the CONSORT 2010 statement. *JAMA*, *330*(21), 2106–2114. https://doi.org/10.1001/jama.2023.19793.

Knowles, M. S. (1988). *The Modern Practice of Adult Education: From Pedagogy to Andragogy*. Cambridge Books.

Ministry of Education Malaysia. (2015). *Malaysia education blueprint 2015–2025 (higher education)*. https://www.mohe.gov.my/en/download/publications-journals-and-reports/pppm-2015-2025-pt/102-malaysia-education-blueprint-2015-2025-higher-education/file

Montgomery, A. A., Peters, T. J., & Little, P. (2003). Design, analysis and presentation of factorial randomised controlled trials. *BMC Med. Res. Methodol.*, *3*, 26. https://doi.org/10.1186/1471-2288-3-26.

Myrsiades, L. (2001). Looking to lead: A case in designing executive education from the inside. *J. Manag. Dev.*, *20*(9), 795–812. https://doi.org/10.1108/EUM0000000006161.

Oreopoulos, P. & Dunn, R. (2013). Information and college access: Evidence from a randomized field experiment. *Scand. J. Econ.*, *115*(1), 3–26. https://doi.org/10.1111/j.1467-9442.2012.01742.x

Schulz, K. F., Altman, D. G., & Moher, D. (2010). CONSORT 2010 statement: Updated guidelines for reporting parallel group randomised trials. *BMJ*, *340*, c332. https://doi.org/10.1136/bmj.c332.

Sen, A. (2001). *Development as Freedom*. Oxford University Press.

Sweller, J. (1988). Cognitive load during problem solving: Effects on learning. *Cogn. Sci.*, *12*(2), 257–285. https://doi.org/10.1207/s15516709cog1202_4.

Tom, A., Senoaji, A., Wydia, C., Aryadewa, E., Azhar, F. S., & Rayindra, H. G. (2026). The effect of blended learning implementation on the academic achievement of Binus University students in the business statistics course. *J. Mod. Educ. Res.*, *1*(1), 16–29.

Vickers, A. J. & Altman, D. G. (2001). Analysing controlled trials with baseline and follow up measurements. *BMJ*, *323*(7321), 1123–1124. https://doi.org/10.1136/bmj.323.7321.1123.

White, I. R., Horton, N. J., Carpenter, J., & Pocock, S. J. (2011). Strategy for intention to treat analysis in randomised trials with missing outcome data. *BMJ*, *342*, d40. https://doi.org/10.1136/bmj.d40.

World Economic Forum. (2024). *The 2020s will be a decade of upskilling. Employers should take notice*. https://www.weforum.org/stories/2024/01/the-2020s-will-be-a-decade-of-upskilling-employers-should-take-notice/